

Characterising the Urban Heat Island of Mumbai

Day–night contrast from Landsat and seasonal variation from MODIS

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Abstract

We measure the surface urban heat island of Mumbai two ways. Part A uses a Landsat 8 day/night pair from 8–9 October 2015 at 30. Part B uses MODIS eight-day composites from 2019–2023 at 1, across four seasons, on both Terra and Aqua, day and night. Both use the same administrative boundary and the same land-cover definition of urban.

Daytime UHI is strongly seasonal and changes sign: Terra MODIS gives -2.72 in pre-monsoon and $+1.67$ in post-monsoon. Night-time UHI is positive in every season, between $+1.38$ and $+2.80$. The Landsat pair gives $+2.39$ by day and $+2.13$ by night once emissivity is corrected, the latter agreeing to within 0.04 with MODIS for the same season and year. The central finding is that Mumbai’s daytime surface heat island is not a fixed property of the city but a seasonal one, driven largely by how hot the surrounding countryside becomes, while the night-time heat island persists year-round.

1 Introduction

Mumbai is among the most densely populated cities in the world, built on a coastal peninsula with limited room to expand outward. Replacing vegetated and bare ground with concrete, asphalt and roofing changes the surface energy balance: built materials store more heat during the day and release it slowly after sunset, while the loss of vegetation removes the evaporative cooling that would otherwise moderate daytime temperatures. The consequences are practical: heat stress and heat-related mortality, higher building cooling demand and electricity consumption, degraded outdoor thermal comfort, and interaction with air pollution.

This report measures the surface urban heat island, not the canopy-layer air-temperature heat island the term often implies. The two are related but not interchangeable: surface temperature responds faster and varies more sharply over short distances, and over built surfaces by day it can exceed air temperature by several degrees. Everything reported here is a radiometric surface measurement.

Two questions are addressed. Part A asks how the heat island differs between day and night, using a Landsat 8 pair acquired one day apart in October 2015 at 30. Part B asks how it varies across seasons, using MODIS eight-day composites from 2019–2023 at 1 on two platforms. Both parts share one boundary and one definition of urban and rural, so that differences between them reflect sensor and season rather than method.

2 Study area and shared definitions

Both parts use the same area of interest and the same urban and rural zones, so that any difference between them comes from the sensor or the season rather than from the method. The one departure is a 100 geometry simplification applied in Part A for computational reasons, sub-pixel at both statistics scales, described in Section 3.1.

2.1 Boundary

Source	Global Administrative Areas (GADM), version 4.1
Level	Level-2 (district)
Unit	NAME_2 = “Mumbai City”, GID_2 = IND.20.17.1
Format	ESRI shapefile
CRS	EPSG:4326, WGS 84 geographic
Area	481.56 (ellipsoidal), 480.10 (file attribute)

Table 1: Boundary provenance.

No reprojection was applied. Earth Engine works in EPSG:4326 internally, and all areas were computed on the ellipsoid rather than in projected units, which is why the two area figures in Table 1 differ slightly. The 1.5 gap is a difference of method, not of data.

A note on the label. GADM names this polygon “Mumbai City” (IND.20.17.1), but its bounding extent runs from 18.891 N to 19.269 N and the mapped area is 480.1. The Mumbai City district proper, the island city, ends near 19.03 N and covers roughly 157. The polygon used here therefore spans both Mumbai City and Mumbai Suburban districts, that is, Greater Mumbai, despite carrying the narrower GADM label. It is referred to as Greater Mumbai throughout this report, and the GADM identifier is given above so the exact polygon can be retrieved.

Why this level, and what the alternatives would have done. Three definitions of “Mumbai” were available. The municipal limit of Greater Mumbai, administered by the MCGM, is the polygon used here. The Mumbai Metropolitan Region (MMR) extends much further, taking in Thane, Kalyan, Navi Mumbai and a large area of open land between them. A satellite-derived built-up extent would be a third option, drawing the boundary from the imagery rather than from administration.

The municipal limit was chosen because it matches the unit that actually governs the surface: building regulation, land use and green space in Greater Mumbai are set by one authority, so the boundary corresponds to a coherent management area rather than an arbitrary line. It is also the unit against which any policy response would be framed.

The alternatives would have moved the result in opposite directions. MMR would pull a great deal of non-urban land inside the boundary (the region is far from uniformly built), diluting the urban mean with vegetated and agricultural pixels and lowering the measured UHI. It would also consume much of the land that here serves as the rural reference, forcing the ring further out. A built-up extent would do the reverse: by excluding parks, water bodies and undeveloped land from the urban zone it would raise the urban mean and increase the measured UHI, at the cost of a boundary that shifts with the classification threshold and has no administrative meaning.

2.2 Urban and rural zones

Zones come from ESA WorldCover v200 (2021) [5], a 10 global product with eleven classes.

- **Urban:** built-up (class 50) inside the boundary. 204.72.
- **Rural:** tree cover, shrubland, grassland, cropland and bare/sparse (classes 10, 20, 30, 40, 60) in a ring from 5 to 20 beyond the boundary. 962.13.

The 5 inner gap keeps the peri-urban fringe and the city’s thermal plume out of the reference. The 20 outer limit keeps the ring in the same coastal climate. Both are varied in Section 4.6.

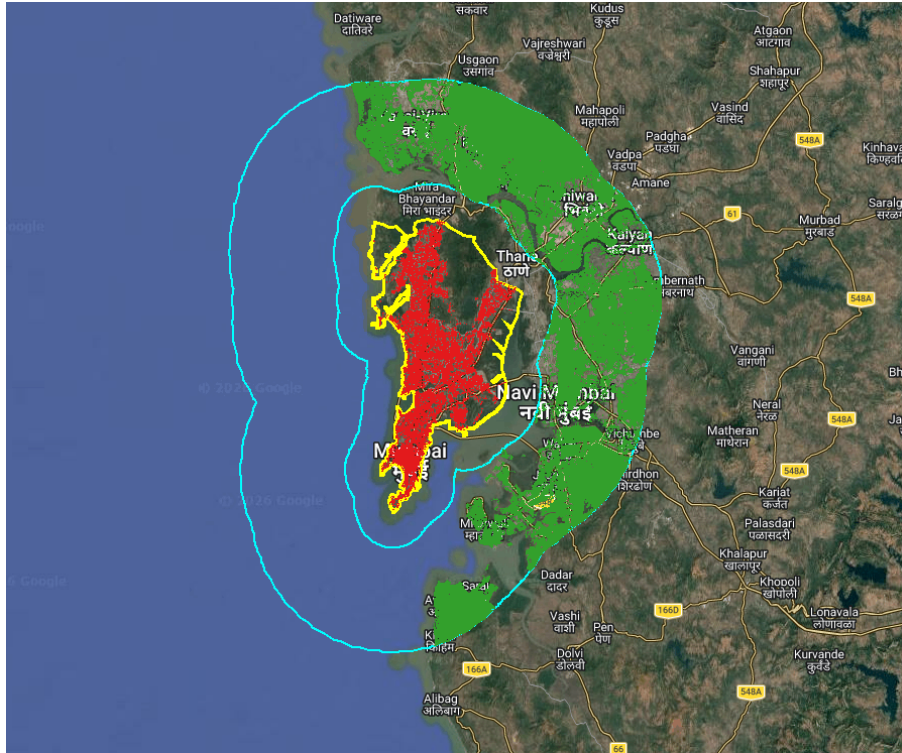


Figure 1: Urban zone (red), rural ring (green), district boundary (yellow), outer edge of the ring (cyan). The western half of the ring is open sea and drops out once water is excluded.

Classes that are neither clearly urban nor clearly rural. Three WorldCover classes are neither clearly urban nor clearly rural. Herbaceous wetland (90) and mangrove (95) were excluded from the rural reference: Mumbai’s creeks are mangrove-lined and tidally inundated, so these surfaces hold standing water for much of the tidal cycle and behave thermally like water rather than land. They were not moved into the urban zone either, since they are not built. Moss and lichen (100) and snow and ice (70) do not occur in the study area. Bare and sparse vegetation (60) was kept in the rural reference: it is genuinely non-built, and excluding it would have biased the reference toward vegetated, and therefore cooler, surfaces.

Mask erosion. No mask erosion (shrinking each zone inward by a fixed radius to discard mixed-pixel edges) was applied to either the urban or the rural mask. A different mechanism was used instead to achieve the same purpose. The urban zone is bounded by the administrative edge rather than eroded, and contamination of the *rural* reference by the urban fringe, the effect erosion is normally used to prevent, is handled by the 5 inner gap of the ring (Section 4.6): rather than eroding a rural mask that touches the city, the ring is defined so that it never touches the city in the first place.

The ring’s inner and outer radii were therefore treated as the parameter to sweep, in place of an erosion radius. The same question the brief asks of erosion, how far the buffer extends and how much UHI moves as a result, is answered by Table 13 and Table 6: moving the inner edge from 2 to 10 shifts MODIS UHI by up to 0.50, and the same sweep on Landsat shifts it by at most 0.18. The urban mask was left un-eroded because it is defined by the administrative boundary rather than by a buffer, so there is no equivalent contamination on that side to remove.

Water and wetland exclusion. Permanent water (class 80) was removed from both zones.

The reason is physical. Water has very high heat capacity and thermal inertia, so its surface temperature barely moves over the diurnal cycle while land swings widely. Leaving it in would

Cover	Area (°)	Share
Permanent water (80)	1534.65	55.1 %
Wetland and mangrove (90, 95)	73.64	2.6 %
Built-up (50)	214.63	7.7 %
Usable rural cover	962.13	34.5 %
Ring total	2785.04	100 %

Table 2: Composition of the 5–20 rural ring. Over half is sea.

anchor the rural mean to a surface that cannot respond to the forcing being measured.

The bias would also change sign between day and night: the sea is cooler than land by day, which would inflate daytime UHI, and warmer by night, which would suppress night-time UHI. Section 4.7 measures this rather than assuming it.

3 Part A: Day and night UHI from Landsat

3.1 Method

	Daytime	Night-time
Scene	LC08_148047_20151008	LC08_018197_20151009
Date	8 October 2015	9 October 2015
Collection	LANDSAT/LC08/C02/T1_L2	LANDSAT/LC08/C02/T2
Level	Level-2 (surface temperature)	Level-1 (radiance)
Band	ST_B10	B10
Cloud cover	8.49 %	not reported (−1)
Sun elevation	57.88	−61.32
Valid-pixel fraction, AOI	88.0 %	100 %
Valid-pixel fraction, ring	95.2 %	99.6 %

Table 3: Landsat scenes used in Part A.

The two scenes are one day apart, which is what makes the day/night comparison meaningful. A larger gap would mix the diurnal signal with changing weather.

Night-time Landsat acquisitions are rare and irregular. They are ascending-node passes, so the same ground location carries a different WRS-2 path/row by night than by day, and they are frequently filed in Tier 2 rather than Tier 1 because of looser geometric accuracy. Both tiers were therefore searched, and because night scenes cannot be chosen freely, the daytime scene was picked *after*: the nearest clear daytime acquisition to the identified night scene, rather than the other way round.

This pair was chosen over the alternatives for three reasons. The two scenes are one day apart, so the diurnal comparison is not confounded by a change in synoptic conditions. The daytime scene has 8.49 % cloud cover and retains ample valid pixels after masking (reported below). And the night-time scene’s temperatures are physically plausible for a Mumbai October night; an earlier candidate pair in November was rejected because its night-time values were anomalously low, in a range inconsistent with the season.

Cloud masking. Both scenes were screened on QA_PIXEL, masking bits 0–5: fill, dilated cloud, cirrus, cloud, cloud shadow and snow. Bit 6 (clear) and bit 7 (water) were left unmasked; water is handled through WorldCover instead, so that the two parts treat it the same way. Valid-pixel fractions after masking are given in Table 3 and discussed in Section 3.2.

From radiance to temperature.

Daytime. The Collection 2 Level-2 product supplies surface temperature directly. `ST_B10` was scaled by 0.00341802 with an offset of 149.0 to give kelvin, then converted to Celsius. The USGS single-channel retrieval behind this product applies its own emissivity correction, derived from ASTER GED.

Night-time. No Level-2 surface temperature is produced for night acquisitions, so the Level-1 product was used. Band 10 digital numbers were converted to spectral radiance using the scene’s own `RADIANCE_MULT_BAND_10` and `RADIANCE_ADD_BAND_10`, then inverted through the Planck function with $K_1 = 774.8853$ and $K_2 = 1321.0789$:

$$T_b = \frac{K_2}{\ln\left(\frac{K_1}{L_\lambda} + 1\right)} \quad (1)$$

The night-time figure is brightness temperature, not LST. No emissivity correction was applied to the night scene, so what Part A reports at night is T_b . Brightness temperature is lower than land surface temperature wherever $\varepsilon < 1$, and the offset differs between built-up ($\varepsilon \approx 0.97$) and vegetated ($\varepsilon \approx 0.98$ – 0.99) surfaces, so it does not cancel in the urban-minus-rural subtraction. This matters when the night figures are compared against MODIS in Section 5, where MODIS night LST *is* emissivity-corrected.

Both brightness temperature and emissivity-corrected LST were computed for the night scene, using a land-cover lookup for emissivity. Table 4 reports the night-time result both ways: 1.15 from brightness temperature and 2.13 from emissivity-corrected LST. The corrected figure is treated as the night-time result throughout this report, and the uncorrected one is retained for comparison in Section 3.3.

A land-cover lookup was used rather than the NDVI threshold method because NDVI requires the red and near-infrared bands, which carry no signal at night. The values assigned follow standard tabulations: 0.970 for built-up, 0.985 for vegetated surfaces, 0.960 for bare ground and 0.990 for water. The daytime scene needs no such treatment, since the Collection 2 Level-2 product applies its own ASTER-GED-derived emissivity during retrieval, so day and night use different emissivity methods, which is a consequence of what each observation makes available rather than a choice between equals.

Statistics scale. The analysis extent, the district plus the 20 rural buffer, covers about 2800, which at the native 30 is roughly 3.1 billion pixels and beyond what Earth Engine will reduce interactively. Zonal statistics were computed at 100, diagnostics at 200–300, and raster exports left at 30. At 100 the Landsat statistics are still ten times finer than the MODIS grid, so the resolution argument in Section 5 is unaffected. The boundary was also simplified to a 100 tolerance before the ring was built; Part B uses the unsimplified polygon, so the two rings differ by at most 100, sub-pixel for both.

3.2 Results

Day and night UHI are close in magnitude: 2.39 and 2.13. Valid-pixel fractions after QA masking were 88.0% over the city and 95.2% over the ring for the day scene, and 100% and 99.6% for the night scene.

The night figure should be read with care. Fmask keys off the reflective bands, which carry no signal on a night pass, so a near-perfect night valid fraction reflects the mask finding nothing to flag rather than a genuinely clear scene.

Variable	Urban ()	Rural ()	UHI ()	n_{urb}	n_{rur}
Day LST (L2 product)	35.45	33.06	+2.39	17 924	91 196
Night T_b (no correction)	21.46	20.32	+1.15	20 506	95 802
Night LST (class ε)	23.48	21.35	+2.13	20 506	95 802

Table 4: Landsat UHI, 8–9 October 2015, external 5–20 rural ring. Statistics at 100.

WorldCover class	Mean day LST ()
Built-up (50)	36.92
Bare / sparse (60)	34.57
Grassland (30)	33.83
Cropland (40)	33.72
Tree cover (10)	32.04
Mangroves (95)	31.20
Wetland (90)	29.72
Water (80)	27.55

Table 5: Mean daytime LST by land-cover class across the full analysis extent, 8 October 2015.

The table does two things. It confirms that built-up is genuinely the warmest class, by 2.35 over the next warmest (bare and sparse vegetation), so the urban signal comes from the built surface itself rather than from the urban zone happening to contain warm pixels of other kinds. And it quantifies what the exclusions in Section 2.2 were worth: water, wetland and mangrove are the three coolest classes, at 27.55, 29.72 and 31.20 against a built-up mean of 36.92. Admitting water to the rural reference would have introduced a surface 9.37 cooler than the urban zone, which is the direct cause of the inflated daytime figure in Table 7.

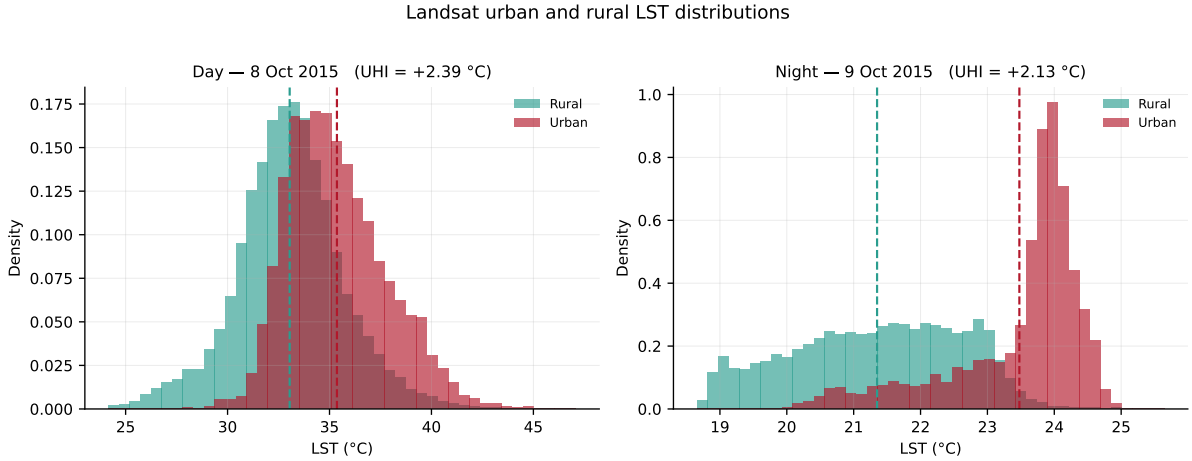


Figure 2: Landsat urban and rural LST distributions. Dashed lines mark zone means. Night uses the emissivity-corrected LST.

The two panels behave quite differently. By day the distributions overlap heavily, with the urban distribution shifted right but sharing most of its range with the rural one. At night the urban distribution collapses into a narrow peak near 24 (standard deviation 1.01 against 1.28 for rural) and sits almost entirely above the rural spread.

The day map shows the built-up core running along the peninsula and into the eastern suburbs consistently warmer than the surrounding green cover, matching the class ranking in

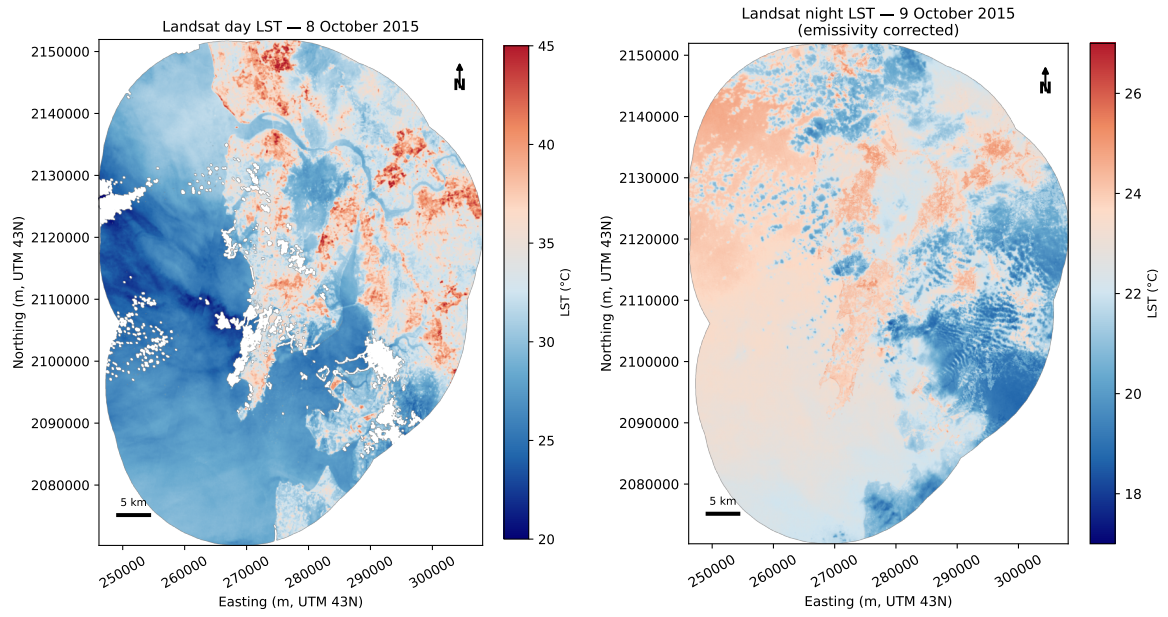


Figure 3: Landsat LST, day (left) and night (right, emissivity corrected). Rasters are exported in UTM Zone 43N; the axes are the projection's easting and northing in metres, not the source boundary's geographic CRS.

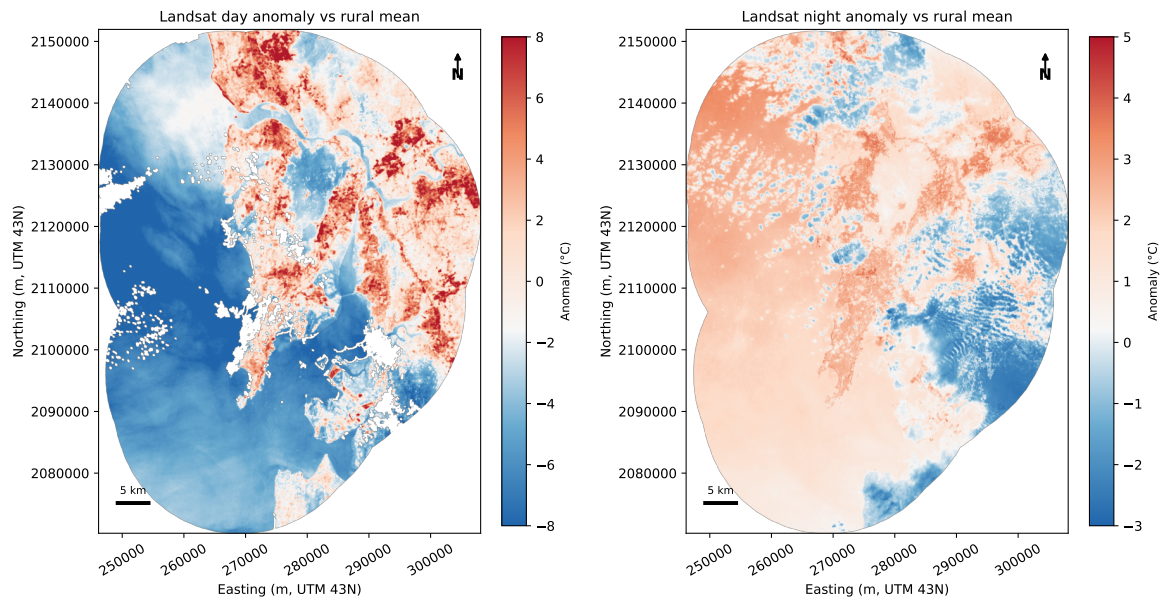


Figure 4: Landsat anomaly against the rural mean, day (left) and night (right). Positive values are warmer than the rural reference.

Table 5. The night map shows a similar but coarser pattern with a visible striped gap in the north-west quadrant, most likely the consequence of the unreliable night-time cloud mask discussed above rather than a real feature of the surface. The anomaly maps make the built-up footprint easier to trace than the raw LST maps do, because they remove the broad north–south gradient present in both scenes.

Day and night UHI are close in magnitude on this pair, 2.39 and 2.13. That similarity is specific to the date rather than general. October falls in the post-monsoon season, and Part B shows this is the one season in which daytime and night-time UHI are both positive and of comparable size (Terra: +1.67 and +2.38). A scene from pre-monsoon would have given a very different daytime answer.

The two measurements also answer different questions. A single scene captures one moment under one set of atmospheric conditions, including whatever soil moisture and cloud history preceded it. A five-year seasonal composite averages that away. Neither is more correct; the Landsat figure is an instantaneous measurement at 30 and the MODIS figure a climatological one at 1.

Against the matched MODIS rows for post-monsoon 2015 (Table 15), the night-time agreement is close: 2.13 against Terra’s 2.09, while the daytime figures diverge, 2.39 against 1.24. The night agreement is the more surprising of the two, since the two retrievals share almost nothing: different sensors, different algorithms, different emissivity treatments, different temporal sampling. Agreement to 0.04 under those conditions suggests the night-time signal is robust to methodological choices in a way the daytime signal is not.

The distributions in Figure 2 show why. At night the urban distribution is a narrow peak (sd 1.01) sitting almost entirely above the rural spread (sd 1.28); nearly every urban pixel is warmer than nearly every rural one. By day the two overlap heavily despite a larger mean difference, so the daytime figure depends much more on which pixels fall in each zone, and therefore on the zone definition, the pixel size, and the mixing that follows from them.

3.3 Sensitivity: rural ring, water, and emissivity

Rural ring. The same four ring variants used in Part B were run on the Landsat scenes.

Ring	Day UHI ()	Night UHI ()
2–20	2.400	2.047
5–20 [†]	2.385	2.127
10–25	2.443	2.223
5–30	2.392	2.129
Range	0.058	0.176

Table 6: Landsat UHI under four rural ring definitions. [†] headline definition.

Landsat UHI moves by at most 0.18 across the four rings, markedly less than the MODIS equivalent in Table 13, which ranges up to 0.50.

Landsat is the less sensitive of the two despite the finer grid, which is at first counter-intuitive. The explanation is temporal rather than spatial. The MODIS sweep operates on a five-year seasonal mean in which the land-cover gradient outside the city (irrigated land close in, drier land further out) is consistently present, so moving the ring systematically changes what the reference is made of. The Landsat sweep operates on one October morning, when that gradient was weaker: post-monsoon soil moisture was still widely distributed, so land at 2 and 10 was thermally more alike than it would be in, say, April.

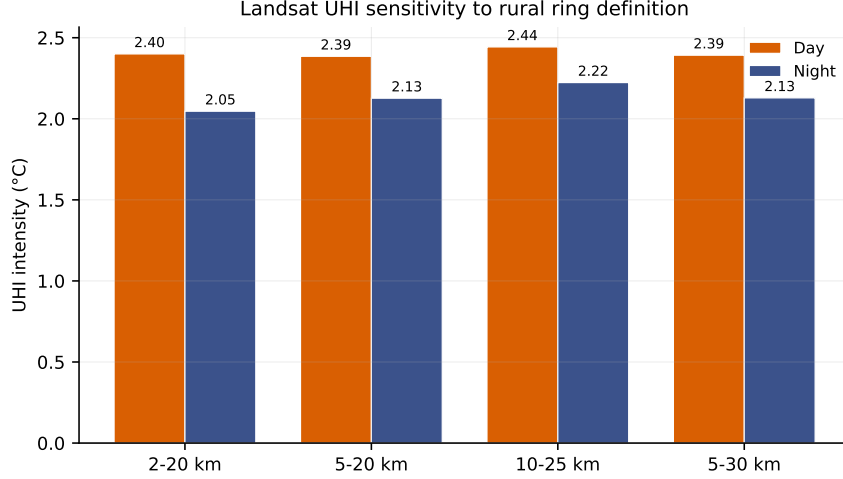


Figure 5: Landsat UHI under four rural ring definitions.

The result is a useful caution. A sensitivity test on a single scene measures how much the answer depends on the zone definition *on that date*, not in general. Part B’s sweep, spanning five years and four seasons, is the better guide to how much the choice matters overall.

Water in the rural reference. The same test run in Part B was repeated on the Landsat scenes.

Mode	Water excl. ()	Water incl. ()	Difference ()
Day	+2.385	+5.635	+3.250
Night	+2.127	+1.200	−0.927

Table 7: Effect of retaining water in the rural reference, Landsat. Rural means with water included: 29.82 by day, 22.28 by night.

Including water more than doubles the daytime figure ($+2.39 \rightarrow +5.64$) and nearly halves the night-time one ($+2.13 \rightarrow +1.20$). The effect is an order of magnitude larger than the MODIS equivalent in Table 14, where the largest shift was 0.45.

The sign reversal matches the prediction and the MODIS result, but the magnitude does not: 3.25 by day here against a MODIS maximum of 0.45.

Two things differ. First, averaging. The MODIS figure is a five-year seasonal mean, and a water surface is almost the same temperature every day of every year, so in a multi-year composite the sea contributes a stable, heavily-averaged value, while the land around it varies and partially averages out. In a single scene the full instantaneous land–sea contrast is present undiluted. On 8 October the daytime mean of water was 27.55 against 36.92 for built-up, a gap of nearly ten degrees (Table 5).

Second, mixing. At 1 a coastal MODIS pixel is part land and part sea, and its retrieved temperature is already a blend; the land–sea boundary is smeared across the pixel grid before any zonal statistic is taken. At 100 the boundary is sharp and sea pixels enter the rural mean at their full, undiluted value. The finer sensor is therefore *more* exposed to the error, not less, which is worth stating plainly, because the intuition that higher resolution is uniformly safer does not hold here.

Emissivity correction. Night-time UHI computed from brightness temperature is 1.15; from emissivity-corrected LST it is 2.13. Applying the correction nearly doubles the result.

The correction does not cancel in the urban-minus-rural subtraction because the two zones have different emissivities: built-up is assigned 0.970 and vegetated surfaces 0.985. A surface with lower emissivity emits less at a given temperature, so correcting for it raises the retrieved temperature more over the city than over the countryside, widening the difference.

The rule of thumb from the module material is roughly 1 of LST error per 1.5 of emissivity error. The built and vegetated values used here differ by 0.015, that is 1.5, so a shift of order 1 between the two zones is exactly what the rule predicts. The observed change in UHI is 0.98, which is close enough to the predicted magnitude to serve as a check that the correction has been applied correctly rather than introducing an artefact.

The report treats the emissivity-corrected value, 2.13, as its night-time result. Brightness temperature is not a surface temperature: it is the temperature a blackbody would need to be to emit the measured radiance, and real surfaces emit less than a blackbody. Reporting T_b as though it were LST would understate the heat island by nearly a degree, and would understate it unevenly, because the error scales with how far each surface’s emissivity sits below unity.

3.4 Reconciling the two parts

An earlier version of Part A used non-built pixels *inside* the city boundary as its rural reference, and did not exclude water. That is a different definition from Part B’s external ring, and it made the two parts non-comparable. Part A was therefore rerun with the external 5–20 ring, water and wetland excluded, so both parts now use identical zones.

The change is large enough to matter:

Variable	Internal ()	External ring ()	Change ()
Day LST	3.811	2.385	−1.426
Night T_b	0.339	1.146	+0.806
Night LST	1.331	2.127	+0.796

Table 8: Effect of moving the rural reference outside the city, with all other settings held fixed. Internal rural means: 31.64 (day LST), 21.12 (T_b), 22.15 (night LST) .

Moving the reference outside the city changes both figures, but in opposite directions: day-time UHI falls by 1.43 while night-time UHI rises by 0.80. A single cause produces both.

An internal reference is made of parks, maidans, mangrove fringe and undeveloped plots embedded within the built fabric. By day these are the coolest surfaces in the city (vegetated, often irrigated, shaded), and are cooler than genuine countryside, which by October is drying out. Using them as the reference therefore exaggerates the daytime contrast. By night the same patches are warmed by the surrounding built fabric and by the urban plume they sit inside, so they are *warmer* than true rural land, and using them suppresses the night-time contrast. The internal reference is biased in whichever direction makes the city look more like its surroundings at night and less like them by day.

The corrected night-time figure, 2.13, agrees with MODIS Terra night for post-monsoon 2015 at 2.09. The two retrievals share almost no methodology (different sensor, resolution, retrieval algorithm and emissivity treatment), so agreement to 0.04 is meaningful corroboration rather than coincidence. The original 0.32 had no such support, and its disagreement with MODIS was the first sign that the reference definition was at fault.

4 Part B: Seasonal UHI from MODIS

4.1 Method

Seasonal LST comes from the MODIS eight-day composites, MODIS/061/MOD11A2 (Terra) [6] and MODIS/061/MYD11A2 (Aqua) [7], both at 1. These are Collection 6.1 split-window retrievals with land-cover-based emissivity.

The eight-day product was chosen over the daily one (MOD11A1/MYD11A1) because the daily product leaves every cloud gap to be handled by hand, and Mumbai has four months of near-continuous monsoon cloud. The eight-day product applies a consistent clear-sky selection instead. The cost is that it hides how few days actually contributed, which is measured in Section 4.3.

Terra and Aqua. Both platforms were used rather than one. Terra crosses at about 10:30 and 22:30 local solar time; Aqua at about 13:30 and 01:30. Terra’s morning pass is close to Landsat 8’s ($\approx 10:30$), so the Terra daytime numbers are the ones comparable with Part A. Aqua sits nearer the daily maximum and the pre-dawn minimum, so it better captures the true extremes. The Terra–Aqua gap within a season isolates time of day from season.

Observation times were read from the `Day_view_time` and `Night_view_time` bands rather than assumed. Over the urban zone they average **10:38** and **22:14** for Terra, and **13:28** and **01:52** for Aqua. Terra’s morning pass is within about eight minutes of Landsat’s.

Season	Months	Role
Winter	January–February	Seasonal contrast
Pre-monsoon (summer)	March–May	Seasonal contrast
Southwest monsoon	June–September	Seasonal contrast
Post-monsoon	October–December	Landsat comparison

Table 9: India Meteorological Department four-season classification [2]. All four were computed.

Season definition. The brief asks for two or three seasons; the first three cover the full range of surface moisture and insolation in Mumbai. Post-monsoon was added because the Landsat pair falls in it, so the cross-sensor comparison in Section 5 has a matching season.

Why IMD and not DJF/MAM/JJA. December–January–February for winter is a mid-latitude convention and does not describe a tropical monsoon coast. Mumbai’s year is set by the onset and withdrawal of the southwest monsoon, not the solstices. December still carries post-monsoon soil moisture and is not thermally like January, which is why IMD places it in post-monsoon. And no IMD season crosses a calendar year, so each year gives exactly one clean instance of each season, which is what makes the year-by-year analysis in Section 4.5 possible. A DJF winter would straddle two years and force an arbitrary choice about each December.

Climatological basis. IMD’s revised normal dates for Mumbai place monsoon onset at 11 June and withdrawal at 8 October [3], which is what fixes the June–September window used for the monsoon season here: it is bounded by the city’s own onset and withdrawal dates rather than a generic national figure. The pre-monsoon window is the driest and hottest part of the year at this station: the 1991–2020 normal gives a May mean daily maximum of 34.3 against 31.2 in January, with December–February rainfall below 2 a month against 384 to 920 a month from June to September [4]. Pre-monsoon (March–May) and winter (January–February) are therefore climatologically distinct: the former is hot and dry, building toward the monsoon onset, and the latter is markedly cooler and equally dry.

Quality screening. Pixels were screened on `QC_Day` and `QC_Night`, keeping mandatory QA (bits 0–1) of 0 or 1 (not cloud-flagged, not otherwise unproduced) and LST error (bits 6–7) of 2 or better, meaning an average retrieval error at or below 3.

The eight-day product is already cloud-screened, so this is not cloud masking. It removes retrievals produced under marginal conditions with large stated errors, which in a monsoon climate cluster in the season of most interest.

A stricter setting (mandatory QA = 0, LST error ≤ 2) was tried first and emptied some night-time zones entirely, producing null zonal means. The relaxed setting was used throughout. Pixel counts never fall below 239 urban and 917 rural.

Compositing. LST bands were scaled by 0.02 to kelvin, then 273.15 subtracted. View-time bands were scaled by 0.1 to local solar hours. For each platform, mode and season, all qualifying composites from 2019–2023 were averaged per pixel. Zonal means were taken at 1000, the native MODIS grid spacing, so no resampling precedes the statistics.

$$\text{UHI} = \overline{T_{\text{urban}}} - \overline{T_{\text{rural}}} \quad (2)$$

The urban zone holds 241 valid pixels at 1 and the rural zone 917–1055. The urban count is the binding constraint on precision.

4.2 Results

Platform	Season	Obs. time	Urban ()	Rural ()	UHI ()	n_{urb}	n_{rur}
<i>Daytime</i>							
Terra	Winter	10:38	32.10	32.50	−0.41	241	1055
Terra	Pre-monsoon	10:37	36.38	39.10	−2.72	241	1055
Terra	Monsoon	10:40	31.16	31.09	+0.07	241	1055
Terra	Post-monsoon	10:32	32.62	30.96	+1.67	241	1055
Aqua	Winter	13:27	35.68	36.55	−0.87	241	1055
Aqua	Pre-monsoon	13:28	39.60	42.24	−2.64	241	1055
Aqua	Monsoon	13:31	32.99	32.07	+0.92	241	1055
Aqua	Post-monsoon	13:32	34.94	33.27	+1.67	241	1055
<i>Night-time</i>							
Terra	Winter	22:14	22.23	19.43	+2.80	239	917
Terra	Pre-monsoon	22:14	26.20	24.30	+1.90	241	977
Terra	Monsoon	22:08	22.78	21.05	+1.73	241	1055
Terra	Post-monsoon	22:09	24.51	22.13	+2.38	241	1042
Aqua	Winter	01:50	21.28	19.03	+2.25	241	1016
Aqua	Pre-monsoon	01:52	25.64	23.11	+2.53	241	1012
Aqua	Monsoon	01:55	22.30	20.92	+1.38	241	1055
Aqua	Post-monsoon	01:53	23.61	21.14	+2.47	241	1055

Table 10: Seasonal surface UHI intensity, 2019–2023 means.

Night-time UHI is positive in every season on both platforms, from +1.38 to +2.80. Daytime UHI is negative in winter (−0.41 Terra, −0.87 Aqua) and strongly negative in pre-monsoon (−2.72, −2.64), near zero in monsoon, and positive only in post-monsoon (+1.67 on both). A negative value means the rural surface is hotter than the urban one: a surface urban *cool* island.

Why daytime UHI is negative in winter and pre-monsoon. The urban means barely move between winter and pre-monsoon (32.10 to 36.38, a rise of 4.28) while the rural means rise much further (32.50 to 39.10, a rise of 6.60). The city has not cooled; the countryside has

Seasonal surface urban heat island intensity, Mumbai (MODIS, 2019–2023)

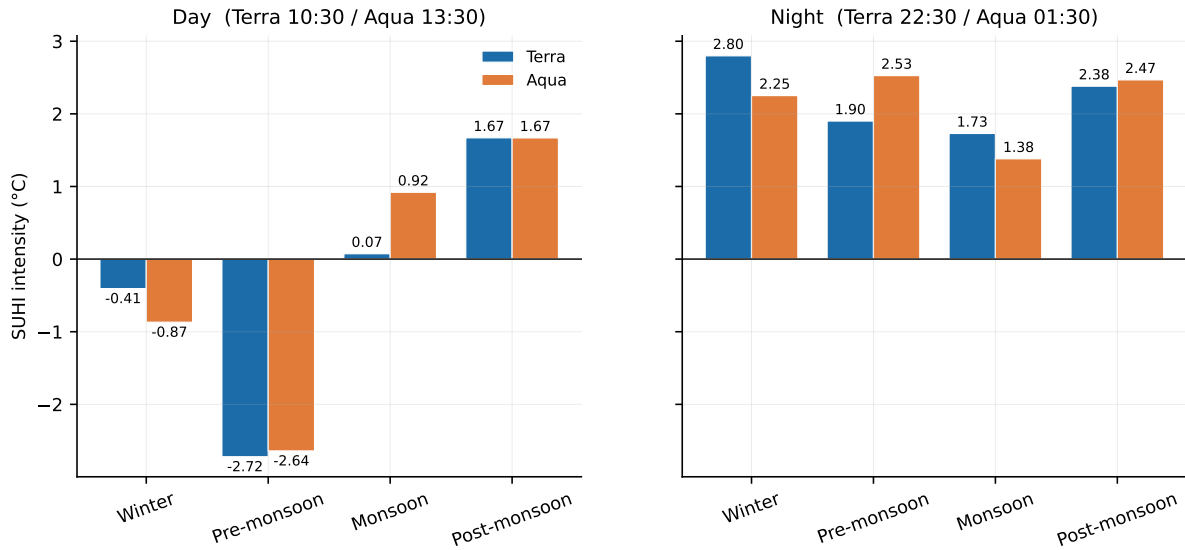


Figure 6: Seasonal SUHI, both platforms, day and night.

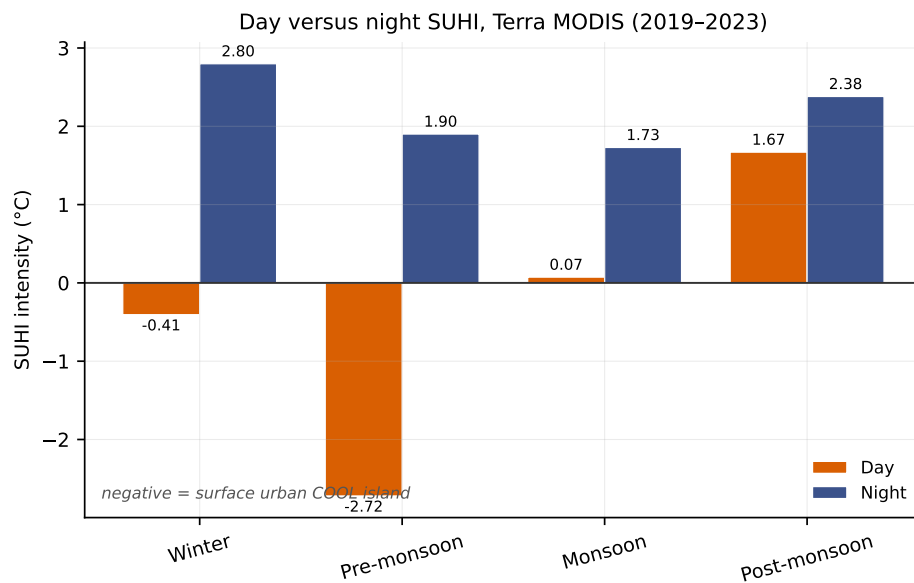


Figure 7: Day and night SUHI side by side, Terra.

heated past it. By the height of the pre-monsoon the rural surface is dry bare soil and senescent vegetation with low thermal inertia and almost no soil moisture left to evaporate, so incoming radiation goes almost entirely into raising the surface temperature. The built surface has higher thermal inertia and conducts heat into the mass of the material below, so for the same radiation load its surface warms less. The result is a surface urban *cool* island of -2.72 .

Why night-time UHI is positive in every season. The same thermal inertia that limits daytime urban warming sustains the night-time heat island. Heat conducted into building fabric and pavement during the day is released slowly after sunset, while the rural surface, having stored little, cools rapidly by radiation. The diurnal ranges make this explicit: the urban zone falls 9.87 between the Terra day and night overpasses in winter against 13.07 for rural, and 10.18 against 14.81 in pre-monsoon. The city is thermally more stable in both directions, slower to heat and slower to cool, which is the definition of higher apparent thermal inertia. This is the nocturnal dominance Oke describes [9], and it holds regardless of what the daytime signal is doing.

Monsoon. Daytime UHI is near zero in monsoon ($+0.07$ Terra, $+0.92$ Aqua). When both surfaces are wet, evaporation dominates the energy balance on both sides of the boundary and the thermal contrast between them largely disappears: the rural surface can no longer run away from the city because its water is not exhausted. This reading should be treated with caution: as Section 4.3 shows, only 13.3 % of monsoon retrievals survive screening, so the composite describes the clearest days of the monsoon rather than the monsoon itself.

Terra against Aqua. The platforms agree closely in some season/mode combinations and diverge in others. The largest daytime gap is monsoon ($+0.07$ against $+0.92$, 0.85 apart) and the largest night-time gap is pre-monsoon ($+1.90$ against $+2.53$, 0.63). Since the only difference between the two is roughly three hours of clock time, these gaps measure how fast the urban–rural contrast is changing at those hours. Aqua’s larger daytime monsoon figure is consistent with its later overpass catching the surface nearer its daily maximum, when the drying of exposed surfaces has had longer to act. The implication for reporting is that a single-platform seasonal UHI is a statement about a particular time of day, not about the season as a whole.

The day–night reversal. Daytime UHI is negative in two of four seasons while night-time UHI is positive in all four. The heat island is therefore not a fixed property of the city: it reverses sign by day depending on how dry the surrounding countryside is, and persists only at night. The two behaviours have the same physical cause, the higher thermal inertia of built surfaces, expressed in opposite directions at opposite ends of the diurnal cycle.

The diurnal ranges confirm this directly. Between the Terra day and night overpasses the urban zone changes by 9.87, 10.18, 8.38 and 8.11 across the four seasons, against 13.07, 14.81, 10.03 and 8.82 for the rural zone. The urban range is narrower in every season, by as much as 4.62 in pre-monsoon. A surface that neither heats nor cools as far as its surroundings will appear cooler than them by day and warmer by night, which is what the seasonal table shows.

4.3 Clear-sky sampling bias

The monsoon composite is not an average over the monsoon. It averages those monsoon days clear enough for a retrieval to be produced and pass screening: the least monsoon-like days of the season.

Monsoon has twice as many composites available as winter but retains the fewest valid retrievals: 10.7 per pixel, 13.3 %, against roughly 38 % elsewhere. By construction these are the

Season	Composites available	Mean valid retrievals	Retention
Winter	40	15.6	38.9 %
Pre-monsoon	55	21.2	38.6 %
Monsoon	80	10.7	13.3 %
Post-monsoon	55	20.4	37.1 %

Table 11: Valid retrievals per pixel against composites available, Terra day, 2019–2023.

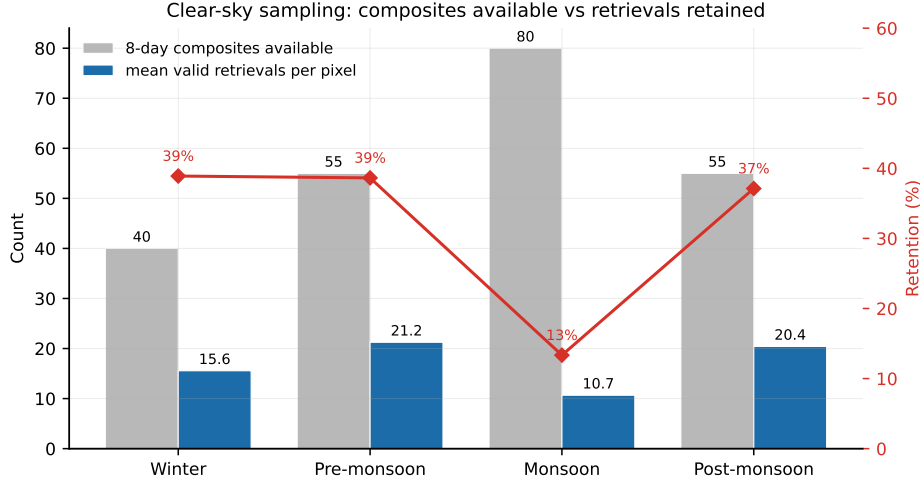


Figure 8: Composites available against retrievals retained. Monsoon has the most input and the fewest usable observations.

least monsoon-like days of the season: the ones with broken cloud, more incoming radiation and drier surfaces than the seasonal norm.

The bias therefore runs toward the pre-monsoon condition. On a genuine monsoon day both surfaces would be wet and the contrast between them smaller still, so the true monsoon UHI is likely nearer zero, or more negative, than the +0.07 retrieved. The retrieved value is an upper bound on the daytime monsoon heat island rather than an estimate of it.

The monsoon row in Table 10 should accordingly carry the least weight of the four. This is not a defect of the product but a limit of thermal infrared remote sensing: the instrument cannot see the surface through cloud, so no amount of compositing recovers the cloudy days. The interannual spread in Section 4.5 shows the same instability from a different angle.

4.4 Urban and rural distributions

The four panels show two different regimes. In pre-monsoon the distributions are almost fully separated, with rural lying to the right of urban: the countryside is not merely hotter on average but hotter pixel for pixel. In post-monsoon the separation reverses and narrows. Winter and monsoon overlap heavily, which is consistent with their near-zero mean differences: the zones shift relative to one another without separating.

The rural distribution is wider in every season, with standard deviations of 1.43–3.02 against 1.24–1.60 for urban. That is expected from the composition of the two zones. The urban zone is a single land-cover class, built-up, and is thermally fairly uniform. The rural ring combines tree cover, shrubland, grassland, cropland and bare ground, each with different moisture and thermal properties, and spans a larger area with more topographic variation. The gap is widest in pre-monsoon (2.60 against 1.60), when differences in soil moisture between rural cover types

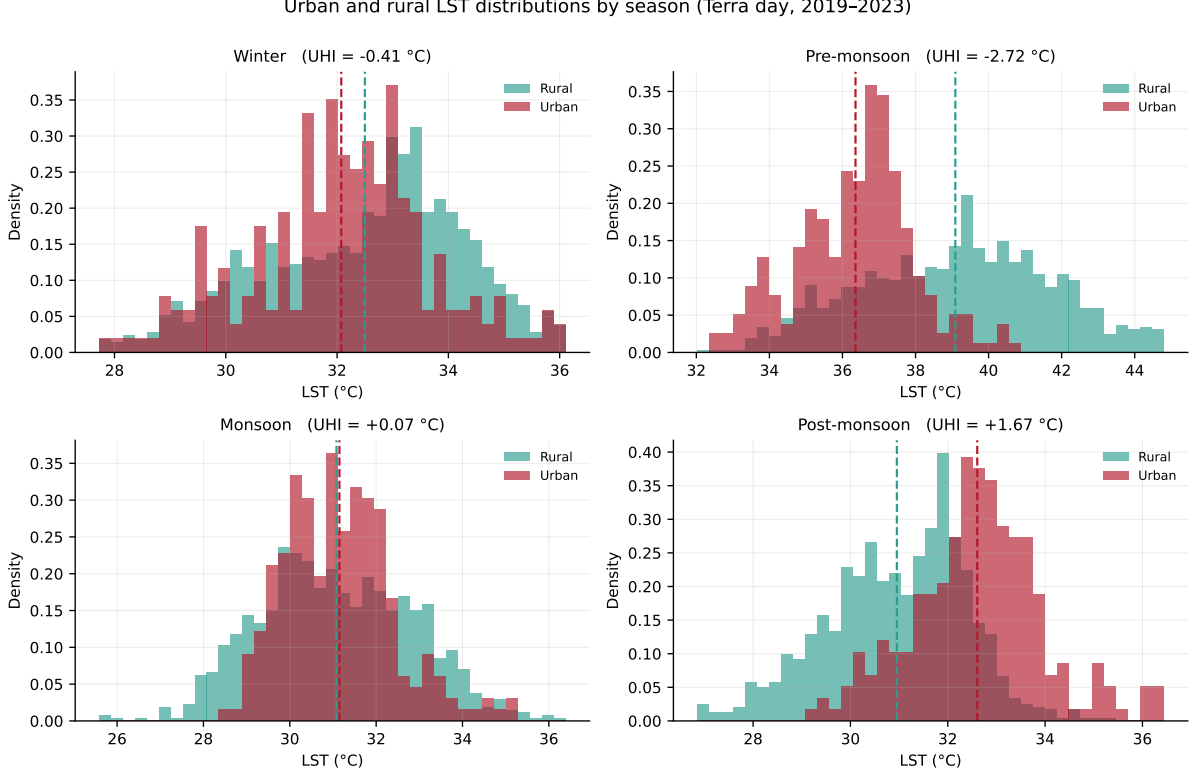


Figure 9: Urban and rural LST distributions by season, Terra day. Dashed lines mark zone means.

are largest.

4.5 Sensitivity: interannual stability

Season	2019	2020	2021	2022	2023	Range
Winter	−1.25	−0.27	−0.38	+0.17	−0.30	1.42
Pre-monsoon	−3.27	−2.92	−2.74	−2.22	−2.45	1.06
Monsoon	+0.06	+1.45	−0.39	−0.31	−0.76	2.20
Post-monsoon	+1.78	+1.59	+2.01	+1.54	+1.41	0.60

Table 12: UHI () by season and year, Terra day.

Pre-monsoon is negative in all five years and post-monsoon positive in all five. Winter changes sign once (2022). Monsoon changes sign three times and has the widest range, 2.20.

The seasonal ordering is preserved in every year. Pre-monsoon is the most negative in all five, post-monsoon the most positive in all five, and winter lies between them in four of five. Only monsoon moves freely, changing sign three times.

Comparing spread against separation makes the point quantitative. Pre-monsoon and post-monsoon are about 4.4 apart while their year-to-year ranges are 1.06 and 0.60: the seasonal difference is four to seven times the interannual noise, so the contrast between them is real. Monsoon has a range of 2.20, larger than its distance from zero in any year, so its sign is not established by this data.

Three of the four seasonal results are therefore robust: the pre-monsoon cool island, the post-monsoon heat island, and the near-neutral winter. The monsoon figure is not, and its instability has the same origin as the sampling bias in Section 4.3: with only 13.3 % retention,

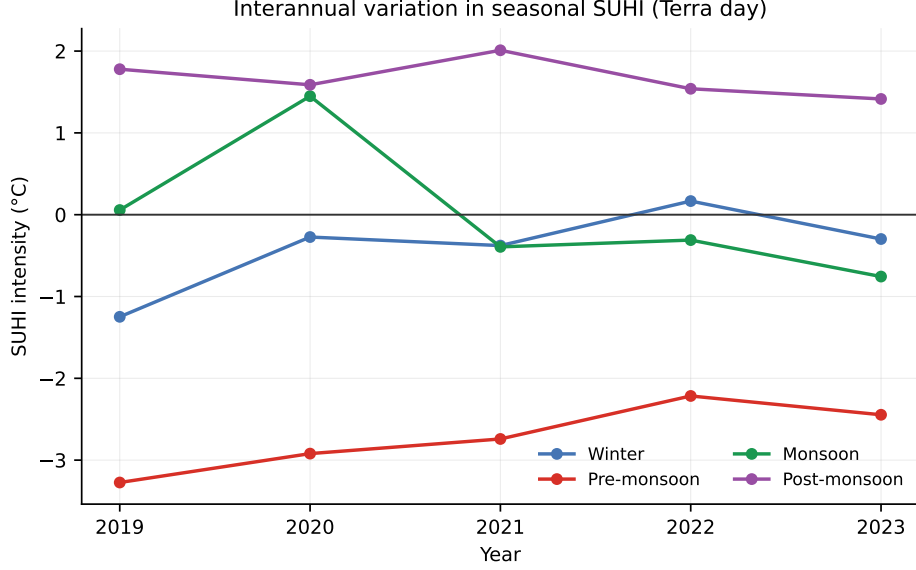


Figure 10: Year-by-year seasonal SUHI, Terra day.

each year’s composite is built from a small and differently-composed set of clear days, so year-to-year variation reflects which days happened to be clear as much as any change in the surface.

4.6 Sensitivity: rural ring definition

Ring	Winter	Pre-mon.	Monsoon	Post-mon.	n_{rur}
2–20	−0.29	−2.55	+0.16	+1.73	1115
5–20 [†]	−0.41	−2.72	+0.07	+1.67	1055
10–25	−0.58	−3.05	−0.05	+1.65	893
5–30	−0.40	−2.74	+0.09	+1.70	1055
Range	0.29	0.50	0.21	0.08	

Table 13: UHI (°C) under four rural ring definitions, Terra day. [†] headline definition.

Moving the inner edge from 2 to 10 shifts UHI by up to 0.50. Pushing the outer edge from 20 to 30 changes almost nothing (0.01 to 0.03).

Moving the inner edge outward makes UHI more negative in every season: winter goes from −0.29 to −0.58, pre-monsoon from −2.55 to −3.05. The rural mean rises as the ring moves away from the city, which means land nearer the boundary is cooler than land further out. Read one way this is the city’s thermal influence reaching past its edge and warming the near-ring; read another, it reflects a land-cover gradient, with more irrigated and vegetated land immediately outside the built-up fringe. Both would produce the same signal, and this data cannot separate them.

Either reading supports keeping a gap between city and reference. The 5 inner edge sits at the point where the trend is already flattening: moving from 2 to 5 changes UHI by 0.17 in pre-monsoon, while 5 to 10 changes it by 0.33, so it excludes the most contaminated land without pushing the reference into a different landscape.

The outer edge barely matters: extending from 20 to 30 moves UHI by at most 0.03. The asymmetry says the useful rural land is concentrated in the inner part of the ring. Land added beyond 20 is either sea, or so similar to what is already included that it does not shift the mean.

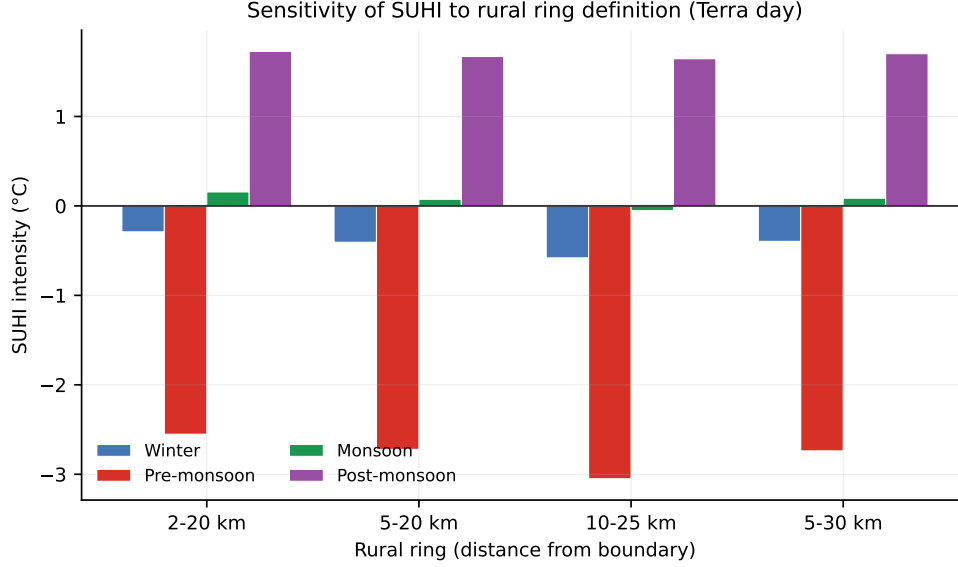


Figure 11: SUHI under four rural ring definitions.

Compared against the Landsat sweep in Table 6, which moves by at most 0.18, MODIS is the more sensitive of the two despite its coarser grid. The likely reason is temporal rather than spatial: the MODIS figure is a five-year seasonal mean in which the land-cover gradient outside the city is consistently present, while the single Landsat scene carries one day's conditions in which that gradient may be weaker.

4.7 Sensitivity: water in the rural reference

UHI was recomputed with water, wetland and mangrove left in the rural reference, everything else fixed.

Mode	Season	Water excl.	Water incl.	Difference
Day	Winter	-0.41	-0.12	+0.291
Day	Pre-monsoon	-2.72	-2.28	+0.445
Day	Monsoon	+0.07	+0.25	+0.176
Day	Post-monsoon	+1.67	+1.77	+0.104
Night	Winter	+2.80	+2.80	-0.004
Night	Pre-monsoon	+1.90	+1.93	+0.031
Night	Monsoon	+1.73	+1.64	-0.087
Night	Post-monsoon	+2.38	+2.35	-0.035

Table 14: Effect of retaining water in the rural reference, Terra. All values .

Every daytime difference is positive (+0.10 to +0.45). Three of four night-time differences are negative (-0.004 to -0.087).

The signs behave as predicted in Section 2.2. Every daytime difference is positive, meaning that including water inflates the apparent heat island, and three of four night-time differences are negative, meaning it suppresses it. The reversal confirms the mechanism: the sea is cooler than land by day and warmer by night, so it pulls the rural mean in opposite directions at the two overpasses.

The daytime effect is about five times the night-time one: up to 0.45 against 0.09. This follows from the diurnal ranges. Water's very high heat capacity means its surface temperature moves only slightly between day and night, while land swings by 9–15. The land–water contrast

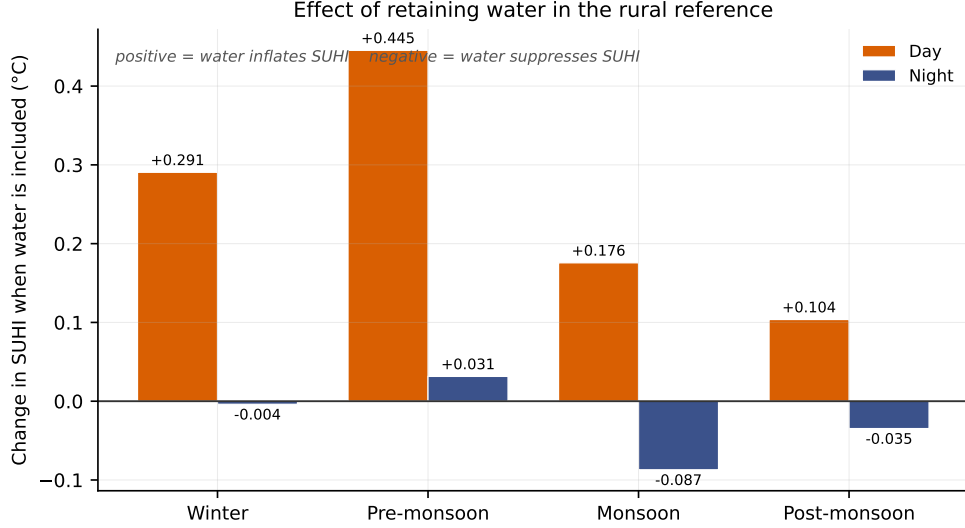


Figure 12: Change in SUHI when water is retained in the rural zone.

is therefore large at the daytime overpass and small at the night-time one, so mixing water into the rural zone displaces the daytime mean much further.

Whether this changes any conclusion depends on the season. The seasonal ordering in Table 10 survives: pre-monsoon remains strongly negative at -2.28 and post-monsoon strongly positive at $+1.77$. But winter shifts from -0.41 to -0.12 , close enough to zero that the sign becomes a matter of measurement error rather than physics. Excluding water is therefore necessary rather than merely tidy, and the necessity is sharpest exactly where the signal is weakest.

5 Comparing Landsat and MODIS

The overlapping season is post-monsoon: the Landsat pair was acquired 8–9 October 2015.

Matching in time. The headline MODIS figures are 2019–2023 means; the Landsat observation is a single day in 2015. Comparing them directly would mix the sensor difference with eight years of urban growth and year-to-year variation. Post-monsoon was therefore recomputed for 2015 alone. Both are given below: the gap between the two MODIS rows is interannual variation, and the gap between the matched MODIS row and the Landsat row is the sensor comparison proper.

5.1 Accounting for the disagreement

The night-time figures agree closely: Landsat 2.13 against Terra 2.09. The daytime figures do not: Landsat 2.39 against Terra 1.24, a gap of 1.15. The question is what explains the daytime gap while leaving the night-time agreement intact.

Resolution. At 1 a MODIS pixel over Mumbai mixes built-up with vegetation, water and open ground, and the retrieved temperature is a blend. Mixing compresses the urban–rural contrast, so MODIS should read lower than Landsat, and it does. The class means in Table 5 give a sense of the scale: built-up at 36.92 against tree cover at 32.04, so a pixel that is half each would read roughly 2.4 below pure built-up. Mixing is therefore capable of explaining a gap of this size on its own.

Sensor	Period	Mode	Obs. time (local)	Res. ($^{\circ}$)	UHI ($^{\circ}$)
Landsat 8	8 Oct 2015	Day	10:30	30	+2.39
Landsat 8	9 Oct 2015	Night	22:00	30	+2.13
Terra	2015	Day	10:41	1000	+1.24
Terra	2019–23	Day	10:32	1000	+1.67
Aqua	2015	Day	13:24	1000	+1.71
Aqua	2019–23	Day	13:32	1000	+1.67
Terra	2015	Night	22:19	1000	+2.09
Terra	2019–23	Night	22:09	1000	+2.38
Aqua	2015	Night	01:49	1000	+2.36
Aqua	2019–23	Night	01:53	1000	+2.47

Table 15: Post-monsoon UHI across both sensors. All rows use the same 5–20 rural ring. The Landsat night value is emissivity-corrected LST. MODIS 2015 rows are matched to the Landsat acquisition year.

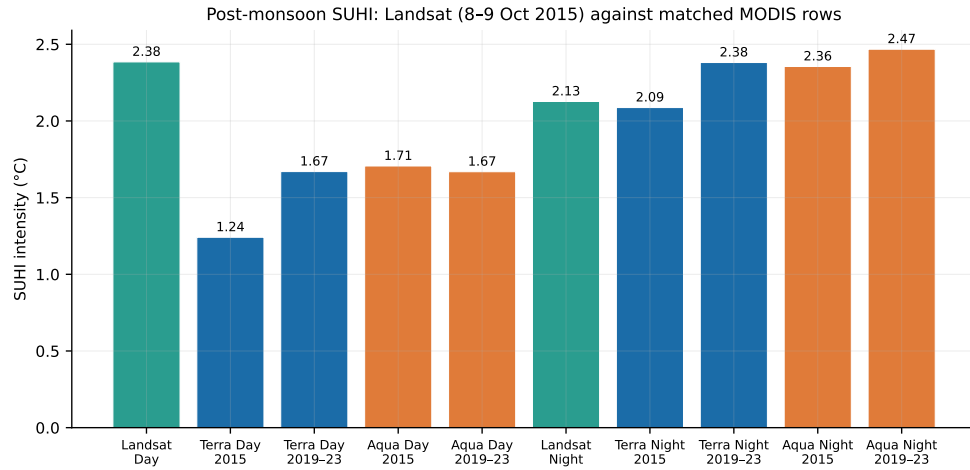


Figure 13: Post-monsoon SUHI: Landsat (8–9 Oct 2015) against MODIS, 2015-only and 2019–2023 mean, both platforms.

Overpass time. Terra’s observed daytime overpass is 10:41 and Landsat’s is near 10:30, about eleven minutes apart. The Terra–Aqua difference for the same season is 0.44 across three hours, so eleven minutes accounts for a few hundredths of a degree at most. Time of day is not the explanation.

Temporal sampling. Landsat is one scene on 8 October; the matched MODIS row is a post-monsoon composite for the whole of 2015, October to December. Even the matched row therefore averages two months the Landsat scene never saw, including December, when the land–city contrast differs from October’s. This is a real contribution and is impossible to remove without a daily product.

Retrieval and emissivity. MOD11A2 uses a generalised split-window with land-cover-class emissivity; Landsat Collection 2 uses a single-channel retrieval with ASTER-GED emissivity. At roughly 1 per 1.5 of emissivity error, systematic differences between the two emissivity sources could contribute a few tenths of a degree.

Which dominates. Resolution is the largest single contributor, with temporal sampling second. The night-time agreement supports this reading: at night the urban distribution is narrow and uniformly elevated (Figure 2), so a mixed pixel averages values that are all similar and mixing costs little. By day the urban distribution is broad and overlaps the rural one, so mixing pulls the urban mean down substantially. The same resolution difference costs little at night and a great deal by day, which is exactly the pattern observed.

6 Limitations

- WorldCover 2021 is used to classify 2015 Landsat observations, so the urban mask is an approximation of the 2015 extent. Mumbai’s built-up extent grew between 2015 and 2021, so some pixels classed as built-up in the 2021 product were not built in 2015. Those pixels would have been cooler by day than genuine 2015 built-up, so the urban mean is biased slightly low and the Part A daytime UHI is, if anything, a slight underestimate.
- The Landsat night-time result relies on a class-based emissivity lookup rather than a per-pixel retrieval, since no reflective-band data are available at night to support the NDVI method used by day.
- Part A rests on one daytime and one night-time scene, so the values carry that day’s weather rather than a climatology.
- Both parts measure surface temperature, not near-surface air temperature. The two differ substantially over built surfaces by day.
- The monsoon composite retains only 13.3% of available retrievals, so it describes the clearest days of the monsoon rather than the season as a whole (Section 4.3). The monsoon figures should be read as an upper bound on the daytime heat island, not an estimate of it.

7 Conclusion

Mumbai’s surface urban heat island is not one number. By day it is strongly seasonal and changes sign: Terra MODIS gives -2.72 in pre-monsoon, when the dry countryside overheats the city, and $+1.67$ in post-monsoon. By night it is positive in every season, between $+1.38$ and $+2.80$. Both behaviours follow from the same property, built surfaces have higher thermal

inertia than their surroundings, expressed at opposite ends of the diurnal cycle. The urban diurnal range is narrower than the rural one in every season, by up to 4.62.

The most robust result is the night-time heat island. It is positive in all four seasons and on both platforms, it survives every sensitivity test applied, and the Landsat and MODIS estimates for the matched season and year agree to 0.04 despite sharing no methodology. The least robust is the monsoon daytime figure, where only 13.3% of retrievals survive screening and the sign changes three times across five years.

The sensitivity tests carry a lesson beyond their individual numbers. Moving the rural reference from inside the city to an external ring changed the night-time result by 0.80 and the daytime result by 1.43. Admitting water to the rural zone changed the Landsat daytime figure by 3.25. Both are larger than the differences between the sensors this report set out to compare. A UHI value quoted without its zone definition is not interpretable.

A follow-up would do three things differently. It would use a daily MODIS product rather than the eight-day composite, so the monsoon sampling bias could be quantified per day rather than inferred from retention rates. It would use a land-cover product contemporaneous with each thermal observation, rather than applying a 2021 classification to 2015 imagery. And it would seek additional night-time Landsat acquisitions, if any exist for Mumbai, since the entire Part A night-time result rests on a single scene whose cloud screening cannot be verified.

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- [9] Oke, T.R., Mills, G., Christen, A. and Voogt, J.A. (2017) *Urban Climates*. Cambridge: Cambridge University Press.

A Earth Engine scripts

Both scripts run top to bottom without editing, and their parameters match those reported above.

Part A: Landsat day/night

<https://code.earthengine.google.com/c1128abfeafdf724eb9a03f539625722>

Part B: MODIS seasonal

<https://code.earthengine.google.com/f00b8b3999cd01e32f97a9d001740569>

B Declaration on the use of AI tools

An AI assistant (Claude, Anthropic) was used during this assignment for the following, and only the following:

- **Earth Engine code.** Drafting and debugging the two analysis scripts, including diagnosing memory-capacity errors and correcting invalid function arguments.
- **Figure generation.** Writing the Python script that renders the charts in this report from the exported CSV files.
- **Report structure and prose.** Producing the L^AT_EX scaffolding, tables and figure placement, and improving the clarity of descriptive passages.
- **Concept explanation.** Clarifying thermal remote sensing concepts covered in the module, such as the behaviour of the MODIS QC bitmask and the form of the single-channel emissivity correction.

It was *not* used to generate the justification of any methodological choice, or the interpretation of any result. Those sections are our own work.

All technical details supplied by the assistant (scale factors, QA bit positions, band names) were verified against the Earth Engine catalogue and the relevant USGS and LP DAAC product guides before being used, and those sources are cited rather than the assistant.